

Matched Filter

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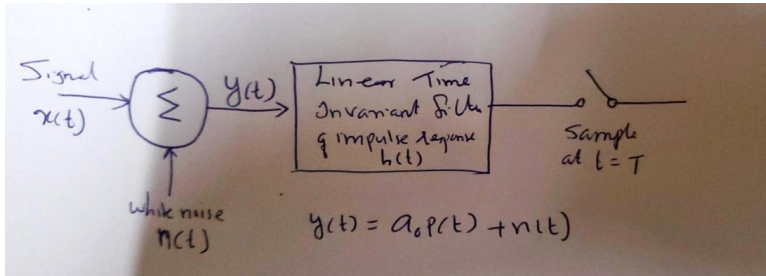
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System Model

Matched Filter

Matched filter is an LTI filter used as an Optimum detector to detect a pulse signal corrupted with AWGN. It is called a Matched filter because its impulse response is matched to the Pulse signal.



- The filter input $y(t)$ consists of a pulse signal $x(t)$ corrupted by AWGN noise $n(t)$

$$\begin{aligned} \text{Transmitted signal } x(t) &= a_0 p(t), \text{ where } p(t) \text{ is the pulse signal} \\ y(t) &= x(t) + n(t) = a_0 p(t) + n(t) \end{aligned} \quad (1)$$

$$\begin{aligned} \text{received signal } r(t) &= y(t) * h(t) \\ &= \int_{-\infty}^{\infty} y(t - \tau) h(\tau) d\tau \\ &= \int_{-\infty}^{\infty} (a_0 p(t - \tau) + n(t - \tau)) h(\tau) d\tau \\ &= a_0 \underbrace{\int_{-\infty}^{\infty} (p(t - \tau) h(\tau) d\tau)}_{\text{signal component}} + a_0 \underbrace{\int_{-\infty}^{\infty} (n(t - \tau) h(\tau) d\tau)}_{\text{noise component}} \end{aligned}$$

Optimal Receiver

Objectives

1. Maximum Signal Power

2. Minimum Noise Power

$\Rightarrow$ SNR at the receiver is Maximum

$$\begin{aligned}\text{Signal Power} &= E(\text{signal component}^2(t)) \\ &= E\left[\left|a_0 \int_{-\infty}^{\infty} (p(t - \tau)h(\tau)d\tau)\right|^2\right] \\ &= E[|a_0|^2]E\left[\left|\int_{-\infty}^{\infty} (p(t - \tau)h(\tau)d\tau)\right|^2\right] \text{ since both are deterministic,} \\ &= P_d \left(\int_{-\infty}^{\infty} p(t - \tau)h(\tau)d\tau\right)^2\end{aligned}$$

• $P_d \Rightarrow$ Power of data symbol

Noise Analysis

- Noise $n(t)$ is AWGN with Zero mean and variance $\frac{N_0}{2}$
- The noise at the output of the filter is:

$$\begin{aligned}\hat{n}(t) &= n(t) * h(t) \\ &= \int_{-\infty}^{\infty} (n(t - \tau)h(\tau)d\tau\end{aligned}$$

$$\begin{aligned}\text{mean } E(\hat{n}(t)) &= E\left[\int_{-\infty}^{\infty} (n(t - \tau)h(\tau)d\tau\right] \\ &= \int_{-\infty}^{\infty} E[(n(t - \tau))]h(\tau)d\tau = 0\end{aligned}$$

$$\begin{aligned}\text{Noise power/variance, } E[\hat{n}(t)^2] &= E[\hat{n}(t) \cdot \hat{n}(t)] \\ &= E\left[\int_{-\infty}^{\infty} (n(t - \tau)h(\tau)d\tau \cdot \int_{-\infty}^{\infty} (n(t - \hat{\tau})h(\hat{\tau})d\hat{\tau}\right]\end{aligned}$$

$$\begin{aligned} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[(n(t - \tau), (n(t - \hat{\tau}))h(\tau)h(\hat{\tau})d\tau d\hat{\tau} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{N_0}{2} \delta(\tau - \hat{\tau})h(\tau)h(\hat{\tau})d\tau d\hat{\tau} \\ &= \int_{-\infty}^{\infty} h(\tau) \left(\int_{-\infty}^{\infty} \frac{N_0}{2} \delta(\tau - \hat{\tau})h(\hat{\tau})d\hat{\tau} \right) d\tau \\ &= \frac{N_0}{2} \int_{-\infty}^{\infty} h(\tau) \cdot h(\tau) d\tau \\ &= \frac{N_0}{2} \int_{-\infty}^{\infty} |h(\tau)|^2 d\tau \end{aligned}$$

- Noise Power at the output = $\frac{N_0}{2} \times \text{Energy of the filter}$

Maximising SNR

Objective

Find $h(t)$ that maximises the SNR at the output

$$SNR = \frac{\text{signal power}}{\text{noise power}} = \frac{P_d \left(\int_{-\infty}^{\infty} p(t - \tau) h(\tau) d\tau \right)^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |h(\tau)|^2 d\tau}$$

Cauchy Schwartz Inequality

$$\left(\int_{-\infty}^{\infty} u(t) v(t) dt \right)^2 \leq \int_{-\infty}^{\infty} u^2(t) dt \cdot \int_{-\infty}^{\infty} v^2(t) dt, \text{ equality for } u(t) = kv(t)$$

- Consider the signal and noise at the output of the sampler ($t=T$):

$$SNR = \frac{P_d \left(\int_{-\infty}^{\infty} p(T - \tau) h(\tau) d\tau \right)^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |h(\tau)|^2 d\tau} \quad (2)$$

- Applying Cauchy Schwartz inequality to equation 2:

$$\begin{aligned}
 SNR &\leq \frac{P_d \left(\int_{-\infty}^{\infty} p^2(T - \tau) d\tau \cdot \int_{-\infty}^{\infty} h^2(\tau) d\tau \right)}{\frac{N_0}{2} \int_{-\infty}^{\infty} h^2(\tau) d\tau} \\
 &= \frac{P_d}{\frac{N_0}{2}} \int_{-\infty}^{\infty} p^2(T - \tau) d\tau
 \end{aligned}$$

Result

SNR is Maximum for equality, ie. when $p(T - \tau) = k h(\tau)$. Take $k=1$.

$\implies$ for matched filter $h(\tau) = p(T - \tau)$

$$(SNR)_{max} = \frac{P_d}{\frac{N_0}{2}} \int_{-\infty}^{\infty} p^2(\tau) d\tau = \frac{P_d}{\frac{N_0}{2}} \int_{-\infty}^{\infty} h^2(\tau) d\tau$$

Assignment

- State and Prove the properties of Matched Filter.
- Analytically prove the equivalence of correlation Receiver and matched filter.
- **Ref:**Digital Communication by Simon Haykins